

26 — Unknown vessel type: L3 falls back to the 500 kW default

Proves · An unrecognised vessel type falling back to the 500 kW default variation.

Mechanics this scenario turns on

- When the Level 3 variation field is empty the backend looks it up from the vessel type (Bulk 250 · Container 1 500 · LNG 1 000 · otherwise the 500 default), all from `appsettings.json`.
- Level 3 (DRC) damps the hotel/mission swing: `variation × 0.8`, minus whatever the battery already shaved, so the same kilowatt is never counted twice.

Panels described below · The input · Why it shares a plant with 19, 25 and 33

Anything not described here — the mechanics above name the step that produced it; `00-ORIENTATION` Part 6 has the full number-to-step index.

Trust · characterisation snapshot, generated from the code. It detects change; it does NOT prove correctness. Figures marked *pending reference verification* have never been checked against anything outside the application.

Read after · scenario 14.

The fourth branch of the variation lookup, and the one a real user hits most often — the vessel catalogue is far larger than the three-entry variation table.

The input

```
vesselTypeName      "Fishing Vessel 800 gt"
hotelLoadVariationKw (omitted)
```

"Fishing Vessel 800 gt" matches no key exactly and contains none of "Bulk Carrier", "Container" or "LNG" as a substring, so `DefaultVesselVariationKw` = **500** applies.

Why it shares a plant with 19, 25 and 33

Identical plant, identical hours, identical fuels — **only the vessel name differs**. That makes the four snapshots directly comparable and turns the lookup into a controlled experiment:

Scenario	vessel name	source	variation	L3
19	Offshore Support	explicit field = 500	500	45.25
26	Fishing Vessel 800 gt	default	500	45.25
33	LNG Carrier 170,000 m³	lookup → LNG	1 000	92.83
25	Container 5,000 TEU	lookup → Container	1 500	142.79

19 and 26 land on the same number by different routes — one from an explicit field, one from the default. The snapshots cannot tell them apart, but the **scenario files** can: 19 sets `hotelLoadVariationKw`, 26 has no such key. That is the distinction under test.

A note on the substring rule. It is generous by design — "Bulk Carrier 10,000 dwt" is meant to match "Bulk Carrier" (scenario 14). The flip side is that a future vessel named, say, "LNG Bunkering Barge" would silently inherit the LNG carrier's 1 000 kW. Worth knowing before adding names.

Takeaway: an unrecognised vessel does not fail and does not warn — it quietly gets 500 kW. That is a reasonable default and an easy thing to forget.